Supplementary Information

Dissecting Sequence, Structure, and Data Recency Effects
in Enzyme Commission Prediction under Temporal Distribution Shift

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S1 Related Work

Classical sequence alignment. Before machine learning, enzyme function annotation relied primarily on sequence similarity transfer. BLAST [1] identifies homologous proteins by computing pairwise alignment scores against curated databases such as UniProtKB/Swiss-Prot; a query enzyme is annotated with the EC number of its closest hit above an e-value threshold. On the EC-Bench temporal test set, BLASTp achieves a weighted F1 of 0.502 at the 100% similarity threshold, dropping to 0.431 when training is restricted to proteins with at most 30% sequence identity to the test set [2]—illustrating the well-known fragility of alignment-based methods for evolutionarily distant queries. Profile-based methods such as PRIAM mitigate this limitation through position-specific scoring matrices derived from multiple sequence alignments, but they cannot integrate non-sequence information such as tertiary structure.

Deep learning for EC prediction. DeepEC [3] and DEEPre introduced 1D CNNs over one-hot amino acid sequences with global average pooling; while effective for EC Levels 1–3, these models struggle at Level 4 due to fine-grained specificity requirements and limited receptive fields. ECPick [4] augments a hierarchical 1D CNN with Dempster–Shafer evidential theory, assigning a trustworthiness score to each prediction. CLEAN [5] reframes EC prediction as metric learning: ESM-1b embeddings are projected into a shared latent space trained with a triplet margin loss. HIT-EC [6] designs a four-level Transformer achieving micro F1 of 0.93 on Swiss-Prot—the current state of the art for sequence-only EC prediction.

Protein language model encoders. ESM-2 [7] is a family of masked language models—scaling from 8M to 15B parameters—pre-trained on over 250 million sequences from UniRef90. The 650M-parameter variant produces per-residue representations of dimension 1,280; the `[CLS]` token embedding encodes a global sequence summary.

Contact map encoding. A protein contact map is a binary symmetric matrix $\mathbf{C} \in \{0, 1\}^{N \times N}$, where $C_{ij} = 1$ if the Cα distance between residues i and j falls below 8 Å [8]. Contact maps encode tertiary fold topology in a translation- and rotation-invariant form. DeepFRI [8] treats the contact map as a sparse adjacency matrix and propagates per-residue features through graph convolutional layers. CLEAN-Contact [9] uses ResNet-50 to encode 2D contact maps, concatenating the resulting representation with ESM-2 features before contrastive learning; on New-392 and Price-149 this achieves a 12% and 16% F1 improvement over sequence-only CLEAN.

EC-Bench temporal evaluation. EC-Bench [2] addresses temporal data leakage by using all reviewed Swiss-Prot entries as of February 2018 for training and 468 proteins added by January 2023 as the test set. The primary metric is the weighted F1-score at Level 4, reported at similarity thresholds of 30%–100%.

S2 Extended Dataset Details

Data splits. We constructed two test sets: (i) *random split* (80/10/10) using a fixed random seed; (ii) *cluster split* using MMSeqs2 [10] at 30% sequence identity, following EC-Bench [2] to evaluate OOD generalization.

Split integrity checks. Because Swiss-Prot contains closely related and sometimes identical protein sequences under different accessions, we audited all train/validation/test partitions for accession and exact-sequence overlap. The random split contains no accession overlap across train, validation, and test sets, but it does contain exact sequence duplicates across partitions; therefore, we treat random-split performance as in-distribution evaluation. For sequence-disjoint evaluation, we use the cluster split, where cluster-train, cluster-validation, and cluster-test contain zero accession overlap and zero exact-sequence overlap. Cluster-test results are reported only for models trained on the corresponding cluster-train split.

Multi-label annotation statistics. EC annotations are naturally multi-label: a single protein may catalyze multiple reactions and therefore receive multiple Level-4 EC numbers. In the main Swiss-Prot training split, 8,385 proteins (3.88%) contain more than one Level-4 EC annotation, with up to nine Level-4 labels for a single protein. The average Level-4 label cardinality is 0.893 per protein when partial EC annotations are included. These statistics motivate sigmoid outputs and binary cross-entropy rather than a single-label softmax classifier.

S3 Training Dynamics

Figure S1 and Table S2 summarize best-checkpoint training dynamics for the main ablation models. All models were trained for up to 30 Phase-1 epochs with frozen ESM-2 representations; Contact-EC-Hier was then trained for an additional 20 Phase-2 epochs with partial unfreezing. B3 (contact map only) converges in approximately 10 epochs, reaching val-hard L4 micro F1 comparable to B1 (0.7650 vs. 0.7655). This fast convergence suggests that the contact-map branch learns a compact structural signal rather than merely memorizing sequence-derived labels. Contact-EC-Hier improves beyond both single-

Table S1: Training hyper-parameters used for the main Contact-EC experiments. Values were fixed before temporal testing unless a split-specific validation threshold is explicitly reported. The 3B and end-to-end variants follow the same evaluation protocol but are single-run analyses due to compute cost.

Parameter	Value
ESM-2 backbone	`facebook/esm2_t33_650M_UR50D` unless noted
ESM-2 dimension	1,280
Contact threshold	8 Å $C\alpha$–$C\alpha$ distance
Contact-map size	256×256
Contact encoder output	512 dimensions
Fusion dimension	1,024 dimensions
Dropout	0.3 for EC-Bench; 0.4 for full/fold-disjoint splits
Batch size	64
Phase 1	30 epochs, frozen ESM-2, learning rate 1.0×10^{-3}
Phase 2	Contact-EC-Hier only: 20 epochs, final two ESM-2 blocks unfrozen, learning rate 1.0×10^{-4}
Optimizer	AdamW
Weight decay	5.0×10^{-4}
Scheduler	CosineAnnealingLR
EC-level loss weights	$(0.1, 0.1, 0.2, 0.6)$ for Levels 1–4
Repeated seeds	42, 43, 44
Checkpoint selection	Validation Level-4 micro F1

modality baselines during Phase 1 and continues to gain after partial unfreezing in Phase 2, reaching 0.8698 on val_hard. The hierarchical sequence-only B2 baseline remains substantially lower throughout training, consistent with the temporal and val-hard underperformance reported in the main text.

Table S2: Best epoch and val-hard L4 micro F1 for each model.

Model	L4 Micro F1 (val hard)	L4 Macro F1 (val hard)	Best Epoch
B1 (ESM-2, flat FC)	0.7655	0.1353	30
B2 (ESM-2, hier. FC)	0.4204	0.0383	30
B3 (Contact map only)	0.7650	0.1335	10
Contact-EC (flat FC, ours)	0.8879	0.2294	29
Contact-EC-Hier	0.8698	0.2141	20 (Phase 2)

S4 Full EC-Bench Leaderboard Comparison

Table S3 reports results on all 468 EC-Bench test proteins, including 309 with partial annotations and 35 with novel EC classes outside the training vocabulary. The all-468 score is reported as an operational closed-set coverage audit, not as a fully defined Level-4 evaluation for partial annotations.

To avoid treating partial EC annotations as Level-4 failures, we additionally evaluate each partial label only up to its deepest annotated hierarchy prefix. For example, EC 3.4.24.- is evaluated through Level 3, whereas EC 1.-.-.- is evaluated only at Level 1. This prefix-level audit is not directly comparable to complete Level-4 F1, but it quantifies whether the model recovers the known functional hierarchy for

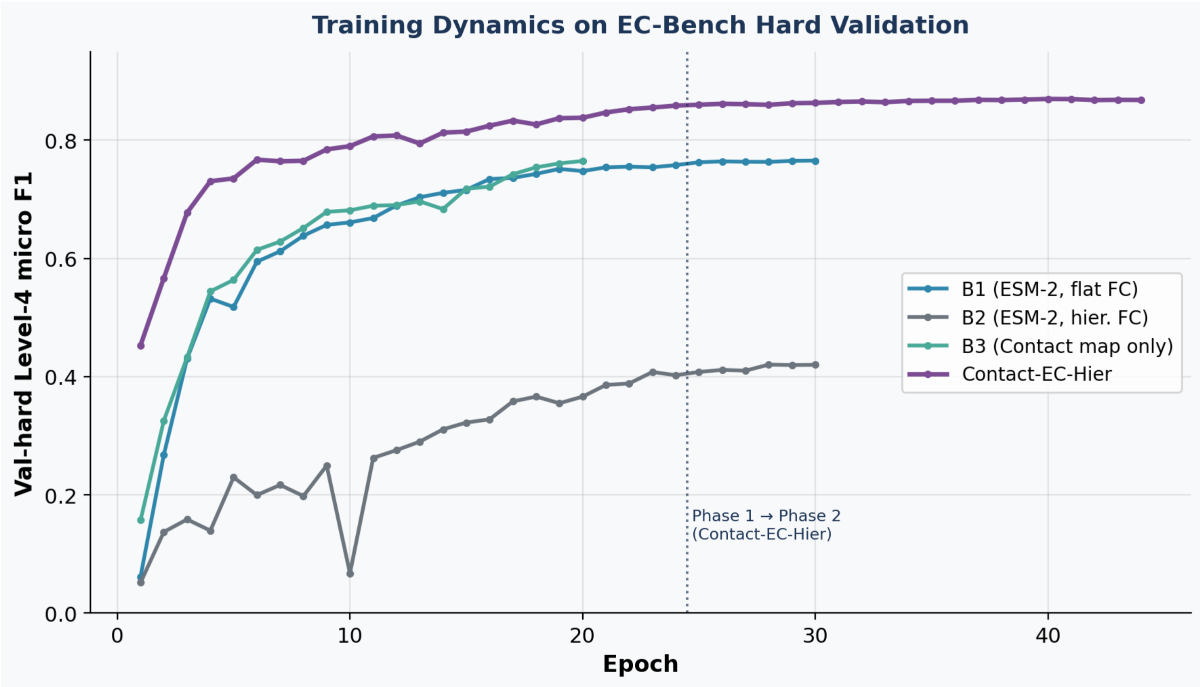

Figure S1: Training dynamics on the EC-Bench hard validation split. Phase 1 uses frozen ESM-2 representations for sequence-containing models; Phase 2 partially unfreezes Contact-EC-Hier. The contact-map-only B3 model rapidly approaches the flat ESM-2 baseline B1, whereas Contact-EC-Hier benefits from sequence–structure fusion and additional Phase 2 optimization.

Table S3: Full EC-Bench temporal test: all 468 proteins vs. evaluable 124. The all-468 columns quantify closed-set coverage under label shift; partial-EC proteins are not fully Level-4 scorable, and novel-EC proteins are outside the SP-2018 vocabulary.

	All 468		**Known-EC 124**	
Model	Micro F1	Weighted F1	Micro F1	Weighted F1
B1 (ESM-2, flat FC)	0.3539	0.3156	0.4216	0.3178
B2 (ESM-2, hier. FC)	0.1098	0.0835	0.1111	0.0835
B3 (Contact map only)	0.3167	0.2625	0.3646	0.2734
Contact-EC (flat FC, ours)	0.3716	**0.4763**	**0.6032**	**0.5026**
Contact-EC-Hier	**0.3771**	0.4325	0.5690	0.4528
EnzBert-learnt†	–	0.463	–	–
Stacked ensemble†	–	0.524	–	–

† EC-Bench leaderboard; weighted F1 on all 468 proteins.

incompletely curated proteins.

Table S4: Hierarchical prefix evaluation on the 309 partial-EC temporal proteins. Partial annotations are scored only at levels where ground truth prefixes are available. Level-4 is omitted because no complete Level-4 ground truth exists for these rows.

Model	L1 Micro F1	L2 Micro F1	L3 Micro F1
B1 (ESM-2, flat FC)	0.8311	0.5805	0.4656
B3 (Contact map only)	0.7930	0.6615	0.5933
Contact-EC (flat FC, ours)	0.8259	0.7322	0.6629
Contact-EC-Hier	**0.8409**	**0.7685**	**0.6975**

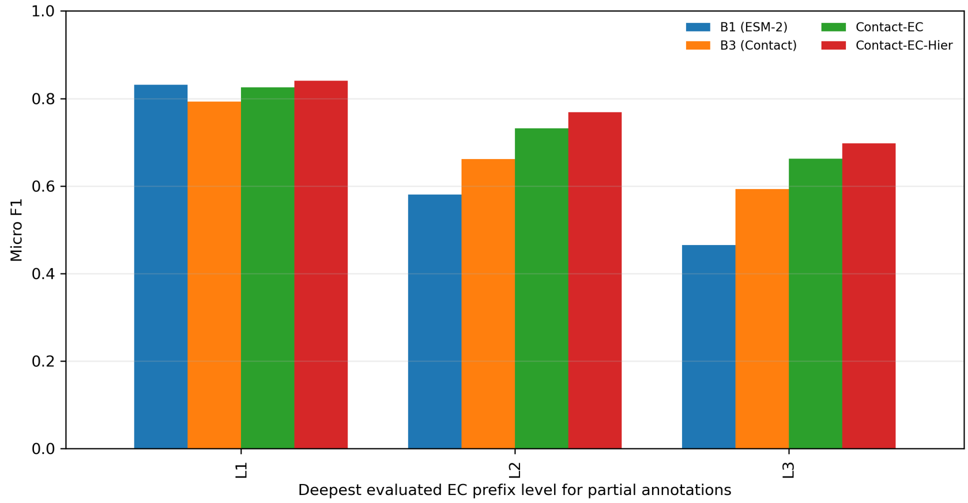

Figure S2: Hierarchical prefix micro F1 on partial-EC temporal proteins. The result shows that partial annotations should be handled as prefix-level supervision rather than assigned Level-4 failures.

Contact-EC (weighted F1 = 0.4763) exceeds EnzBert-learnt (+1.3 pp) despite training on an earlier corpus with a 4.5× smaller backbone. Contact-EC-Hier (0.4325) falls 3.8 pp below EnzBert-learnt; the gap to Contact-EC (+4.4 pp) reflects the flat FC head's superior calibration on partial/novel-EC proteins under distribution shift.

The case-wise comparison confirms that HIT-EC is the stronger temporal predictor on the SP-2023-01 complete known-EC holdout. Contact-EC uniquely recovers two cases: A0A3G9HRC2, a multi-label oxidoreductase with EC 1.14.14.1 and EC 1.6.2.4, and A0A6C0WW38, where HIT-EC predicts the neighbouring methyltransferase class EC 2.1.1.159 instead of EC 2.1.1.160. Conversely, many HIT-EC-only successes are near misses for Contact-EC: for C0HLB8 (EC 3.1.1.4), Contact-EC assigns EC 3.1.1.4 the highest probability but below the fixed 0.5 threshold (0.4931). These cases support the main text's interpretation that Contact-EC is useful for decomposing structure and calibration effects, whereas HIT-EC remains the stronger closed-set classifier on this shorter temporal holdout.

Table S5: Case-wise HIT-EC vs. Contact-EC comparison on the 124 complete known-EC Swiss-Prot 2023-01 temporal proteins. A case is counted as correct if the model recovers at least one true Level-4 EC label.

True L1	Both correct	HIT-EC only	Contact-EC only	Both wrong	Total
1	12	5	1	0	18
2	13	8	1	3	25
3	29	11	0	2	42
4	5	20	0	7	32
5	4	1	0	2	7
All	63	45	2	14	124

EC Level-1 family analysis. To add biological granularity, we recomputed temporal predictions for the ESM-2-only B1 model, the contact-only B3 model, and Contact-EC on the 124 fully evaluable temporal proteins, then stratified Level-4 micro F1 by true EC Level 1 family. Table S6 and Figure S3 show that Contact-EC yields the largest gains over ESM-2 in oxidoreductases (EC 1; +0.6270 micro F1) and hydrolases (EC 3; +0.2753), whereas lyases (EC 4) remain difficult for all Contact-EC variants despite HIT-EC's stronger performance. These results suggest that the utility of contact-map fusion is enzyme-family dependent rather than uniform across EC space. Because several families contain few temporal examples, the analysis should be interpreted as hypothesis-generating rather than mechanistic evidence about active-site geometry.

Table S6: Level-1 family analysis on the 124 complete known-EC Swiss-Prot 2023-01 temporal proteins. Values are Level-4 micro F1 within each true Level-1 family. EC 6 and EC 7 are absent from this evaluable subset.

EC	Family	N	ESM-2	Contact	Contact-EC	HIT-EC
1	Oxidoreductases	18	0.0909	0.2222	0.7179	0.9231
2	Transferases	25	0.6667	0.6154	0.7059	0.8649
3	Hydrolases	42	0.5085	0.4561	0.7838	0.9545
4	Lyases	32	0.1923	0.0833	0.1695	0.6667
5	Isomerases	7	0.3636	0.2500	0.6667	0.7143

Table S7: Contact-EC threshold sensitivity on the 124 complete known-EC Swiss-Prot 2023-01 temporal proteins. The fixed threshold of 0.5 is used in the main benchmark table.

Setting	Threshold	Micro F1	Weighted F1	Precision	Recall	Empty
Fixed global	0.50	0.6032	0.5026	0.7677	0.4967	45
Best global by micro F1	0.65	0.6167	0.4895	0.8506	0.4837	54
0.5 or top-1 if empty	0.50	0.5589	0.5413	0.5764	0.5425	0
Top-1 only	–	0.5126	0.4628	0.5726	0.4641	0

Threshold sensitivity analysis shows that the fixed 0.5 cutoff is not the sole explanation for Contact-

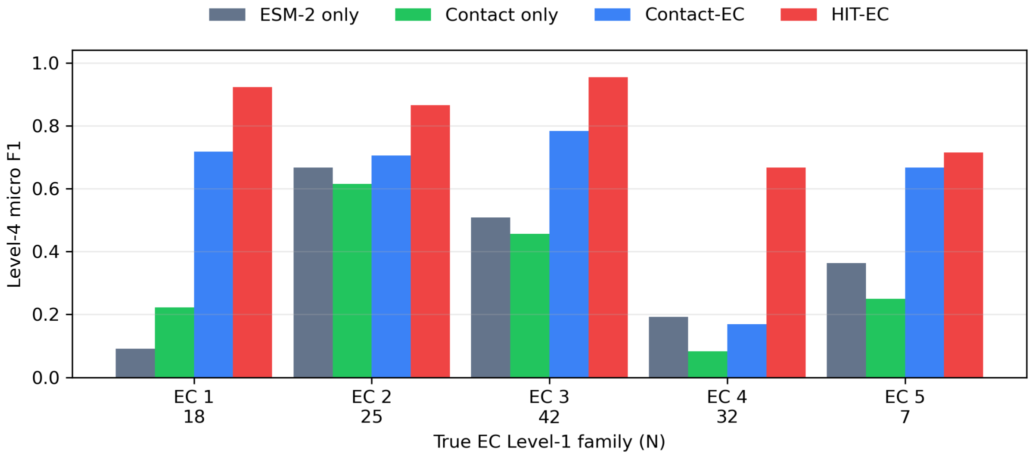

Figure S3: Temporal Level-4 micro F1 stratified by true EC Level-1 family. The largest Contact-EC gains occur in oxidoreductases and hydrolases, while lyases remain a failure mode for the current contact-fusion model.

EC's temporal gap. Raising the global threshold to 0.65 slightly improves micro F1 from 0.6032 to 0.6167 by increasing precision, but reduces recall and weighted F1. Lower-threshold or top-1 fallback variants recover more labels but introduce additional false positives. This behaviour is consistent with a calibration–recall trade-off rather than a simple thresholding artifact.

S5 External Benchmark Evaluation

Table S8: External benchmark results (per-protein F1, threshold = 0.5). CLEAN / CLEAN-Contact results from Ayres et al. [9]. ¶: Contact-EC-ExpA on full SP-2026 (243K); New-392 pre-dates SP-2026, creating likely train/test overlap—interpret as upper bound.

Method	**Backbone**	**Train data**	**New-392**	**Price-149**
CLEAN-Contact [9]	ESM-2 3B	SP-full	**0.566**	**0.525**
CLEAN [5]	ESM-2 3B	SP-full	0.504	0.452
ProteInfer [11]	CNN	UniRef90	0.309	–
B1 (ESM-2)	ESM-2 650M	SP-2018	0.170	0.026
B3 (Contact)	ResNet-50	SP-2018	0.154	0.000
Contact-EC (flat FC)	ESM-2 650M + ResNet-50	SP-2018	**0.3717**	**0.1775**
Contact-EC-Hier	ESM-2 650M + ResNet-50	SP-2018	0.358	0.096
Contact-EC-ExpA¶	ESM-2 650M + ResNet-50	SP-2026	0.7680¶	**0.2161**

New-392. Contact-EC (SP-2018) achieves per-protein F1 = 0.3717, above ProteInfer (0.309) but below CLEAN (0.504) and CLEAN-Contact (0.566). The gap has two causes: (i) CLEAN-Contact uses ESM-2 3B (4.5× larger) and (ii) contrastive retrieval benefits from training-example similarity at inference. Contact-EC-ExpA (SP-2026) achieves 0.7680 but cannot be treated as a fair comparison since New-392 proteins pre-date 2026 and likely appear in the training corpus.

Price-149. Contact-EC (0.2500 micro F1) achieves the highest score among SP-2018 models, but the external Price-149 benchmark remains a major failure mode relative to CLEAN-Contact (0.525). To determine whether this gap is caused by missing inputs, we audited the raw 149 proteins, encoded SP-2018 evaluation subset, label vocabulary, and structure assets. Table S9 shows that input coverage is not the primary explanation: ESM-2 embeddings and nonzero contact maps are available for all 149 raw proteins, and 139 proteins have complete Level-4 EC labels contained in the SP-2018 vocabulary. Instead, the failure is concentrated at exact Level-4 specificity. In the hierarchical diagnostic, Contact-EC retains high coarse EC performance on Price-149 (Level 1 micro F1 = 0.9379; Level 2 = 0.7862; Level 3 = 0.7655), but Level 4 micro F1 falls to 0.0508 (Figure S4). The contact-only B3 model scores 0.0000 on the canonical Price-149 evaluation, and ESMFold-derived contact maps do not recover the gap (Contact-EC ESMFold-map micro F1 = 0.1218). We therefore interpret Price-149 as an external bacterial RefSeq distribution-shift stress test: the model can often identify the broad enzyme family, but its current SP-2018 contact-fusion calibration does not transfer reliably to exact EC subclasses.

Table S9: Price-149 failure audit. Coverage is computed on the raw Price-149 file ($N = 149$); canonical Level-4 model metrics use the encoded SP-2018 evaluation subset ($N = 136$).

Audit item	Count/value	Interpretation
Raw Price-149 proteins	149	external bacterial RefSeq benchmark
Encoded SP-2018 subset	136	canonical flat Level-4 evaluation subset
Proteins with embeddings	149	no missing ESM-2 input bottleneck
Proteins with nonzero contact maps	149	no missing contact-map bottleneck
Raw proteins with all EC labels in SP-2018 L4 vocabulary	139	moderate label-vocabulary coverage
Unique raw Level-4 EC classes	56	external label diversity
Unique raw Level-4 EC classes outside SP-2018 vocabulary	5	limited closed-set mismatch
B1 ESM-2 Level-4 micro F1	0.0324	sequence-only transfer collapses
B3 contact-only Level-4 micro F1	0.0000	contact-only transfer collapses
Contact-EC flat Level-4 micro F1	0.2500	fusion improves but remains below CLEAN-Contact
Contact-EC hierarchical Level-1 micro F1	0.9379	broad enzyme family often recovered
Contact-EC hierarchical Level-4 micro F1	0.0508	exact subclass specificity fails

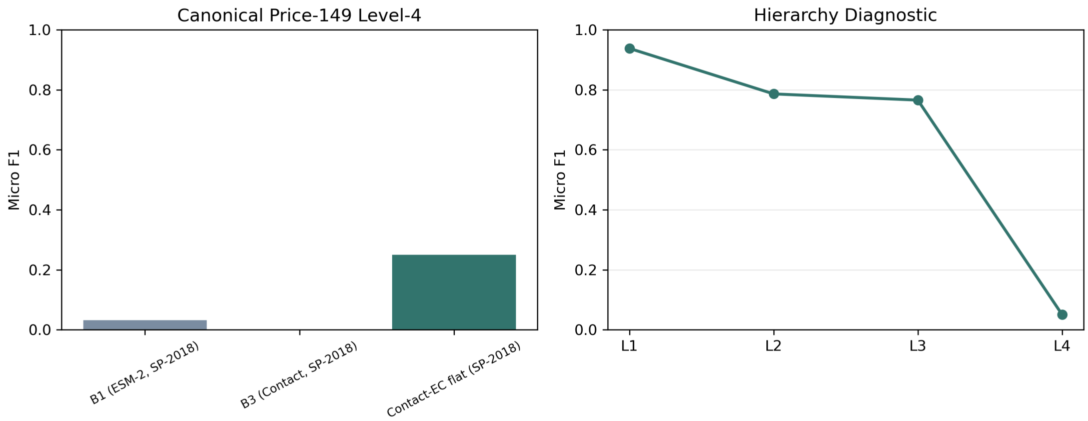

Figure S4: Price-149 failure analysis. Left: canonical Level-4 micro F1 for SP-2018 models. Right: hierarchical diagnostic for Contact-EC, showing that coarse EC levels remain predictable while exact Level-4 specificity collapses.

Contact-EC calibration and prediction-cardinality audit. We further audited whether the Price-149 failure is explained by empty predictions at the fixed 0.5 threshold. Table S10 and Figure S5 show that this is not the only explanation. On Price-149, Contact-EC emits nonempty predictions for 51.5% of encoded proteins, with median top-1 probability 0.5527, but the top-1 hit rate is only 27.9% and Level-4 micro F1 is 0.2500. Thus the external failure reflects exact-subclass specificity and calibration under distribution shift, not simply missing structures, absent embeddings, or universally empty prediction sets. For partial and novel temporal proteins, Level-4 F1 and top-1 hit rate are undefined because the closed-set target is incomplete or outside the training vocabulary; for those groups we report only prediction burden and confidence diagnostics. The flattened multilabel calibration errors are therefore used only as diagnostics because they are dominated by the many negative Level-4 labels.

Table S10: Contact-EC prediction-cardinality and confidence audit at threshold 0.5. Partial and novel temporal proteins are not fully Level-4 scorable, so their exact-hit and micro-F1 fields are reported as N/A.

Dataset	N	Avg. L4 preds	Empty rate	Mean top-1 p	Top-1 hit	L4 micro F1
Temporal known EC	124	0.7984	36.3%	0.6657	57.3%	0.6032
Temporal partial EC	309	0.4595	64.4%	0.3862	N/A	N/A
Temporal novel EC	35	0.4286	74.3%	0.3404	N/A	N/A
Price-149	136	0.6471	48.5%	0.5187	27.9%	0.2500

Late probability fusion baselines. To test whether Contact-EC's gain could be reproduced by simple post-hoc combination of independently trained sequence-only and contact-only models, we evaluated probability-level fusion baselines using B1 and B3 predictions: mean posterior averaging, element-wise posterior maximum, and two fixed weighted averages. These baselines require no retraining and therefore

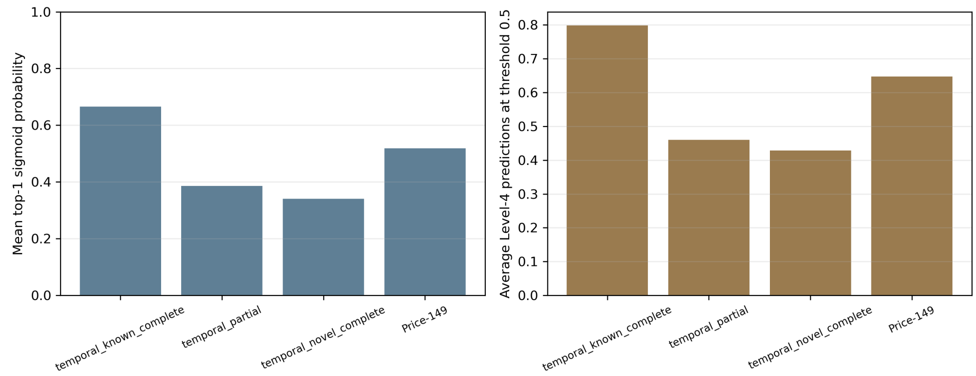

Figure S5: Contact-EC calibration and prediction-cardinality diagnostics. The audit separates empty prediction rate, average predicted-label cardinality, top-1 confidence, and top-1 correctness for temporal known, temporal partial, temporal novel, and Price-149 proteins.

provide a conservative diagnostic rather than a full architectural ablation. As shown in Table S11 and Figure S6, learned Contact-EC substantially outperforms all late-fusion variants on the 124 complete temporal proteins (0.6032 vs. 0.4673 for the best late-fusion baseline). The same pattern holds on Price-149, where late fusion remains near the weak ESM-only baseline while Contact-EC reaches 0.2500. Thus the observed sequence–structure gain is not explained by naive posterior averaging or maximum pooling.

Table S11: Late probability-fusion baselines using independently trained B1 (ESM-2) and B3 (contact-only) models. Values are Level-4 micro F1 at threshold 0.5.

Model	Swiss-Prot 2023 known EC ($N = 124$)	Price-149 ($N = 136$)
B1 (ESM-2)	0.4216	0.0324
B3 (Contact)	0.3646	0.0000
Late mean B1+B3	0.3770	0.0118
Late max B1+B3	0.4673	0.0312
Late weighted 0.7B1+0.3B3	0.4322	0.0231
Late weighted 0.3B1+0.7B3	0.3598	0.0000
Contact-EC learned fusion	**0.6032**	**0.2500**

Trainable fusion architecture controls. We also trained three same-encoder, flat-head sequence–structure fusion controls on the SP-2018 EC-Bench training corpus and evaluated them on the complete Swiss-Prot 2023-01 temporal subset. The concat control concatenates projected ESM-2 and contact-map features before an MLP, the sum control adds the two projected features, and the gated-MLP control learns a feature-wise interpolation between the two projections. Unlike the late-fusion baselines, these controls are trained end to end over the downstream fusion layers and therefore directly test whether Contact-EC's gated additive fusion is merely a generic benefit of adding contact-map features. Table S12 shows that Contact-EC remains strongest across three seeds, although concat fusion is close enough that the architecture should be interpreted as a useful controlled fusion choice rather than a large standalone architectural breakthrough.

Table S12: Trainable simple fusion baselines on the complete Swiss-Prot 2023-01 temporal subset ($N = 124$). Values are mean $\pm$ sample standard deviation across seeds 42/43/44. All trainable controls use the same contact encoder family and flat Level-4 prediction head as Contact-EC.

Model	Micro F1	Weighted F1	Precision	Recall	Rare-EC F1
Concat flat FC	0.5922 ± 0.0150	0.4777 ± 0.0116	0.8269 ± 0.0479	0.4619 ± 0.0136	0.4729 ± 0.0113
Sum flat FC	0.5505 ± 0.0230	0.4818 ± 0.0067	0.6638 ± 0.0433	0.4706 ± 0.0131	0.3970 ± 0.0112
Gated MLP flat FC	0.5543 ± 0.0525	0.4532 ± 0.0413	0.7768 ± 0.0692	0.4314 ± 0.0458	0.4132 ± 0.0368
Contact-EC flat FC	$\mathbf{0.6241 \pm 0.0170}$	$\mathbf{0.5278 \pm 0.0222}$	0.7997 ± 0.0226	$\mathbf{0.5120 \pm 0.0200}$	$\mathbf{0.5257 \pm 0.0313}$

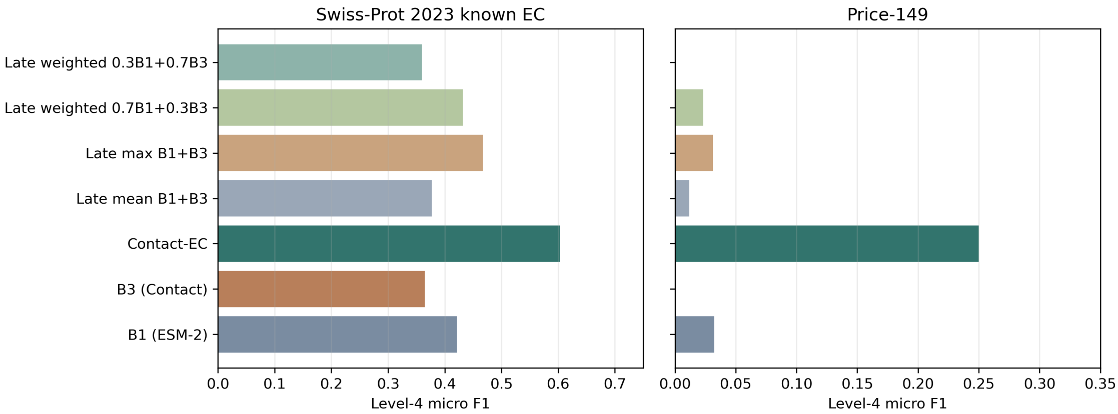

Figure S6: Late probability-fusion baselines. Learned Contact-EC exceeds simple B1/B3 posterior averaging, posterior maximum, and fixed weighted averages on both the temporal known-EC subset and Price-149.

S6 Contact Map Channel Ablation

Table S13: Contact map channel perturbation (B3 checkpoint, random-split test, $N = 22{,}018$ complete-L4 proteins). Inference-time zeroing of channels 1–2 is a pessimistic lower bound on the training-time 1-channel ablation.

Input channels	**L4 Micro F1**	**Δ vs. 3ch**
3-channel (all / short / long)	**0.8684**	—
1-channel† (ch0 only, ch1–2 zeroed)	0.1187	-74.97 pp

We encode contact maps as three channels—all contacts, short-range ($|i-j| < 12$), and long-range ($|i-j| \geq 12$). Inference-time zeroing of channels 1–2 drops L4 micro F1 from 0.8684 to 0.1187 (-74.97 pp), confirming the ResNet encoder exploits the channel decomposition to distinguish secondary from tertiary structure.

S7 Grad-CAM Structural Interpretability

We apply Gradient-weighted Class Activation Mapping (Grad-CAM) [12] to the ResNet-50 contact map branch to verify that Contact-EC encodes structurally meaningful contact patterns rather than superficial statistics.

For **Lysozyme C** (EC 3.2.1.17; UniProt C0HLB7), Grad-CAM highlights two discrete regions centred around residues 85–140 and 180–220, corresponding to the catalytic β-sheet strand and α-helix cluster adjacent to the Glu35–Asp52 active-site dyad [13].

For **Phospholipase A1** (EC 3.1.1.32; UniProt P0DPT0), a single hot spot emerges around residues 110–175, co-localising with the serine–histidine–aspartate catalytic triad of the α/β hydrolase fold [14].

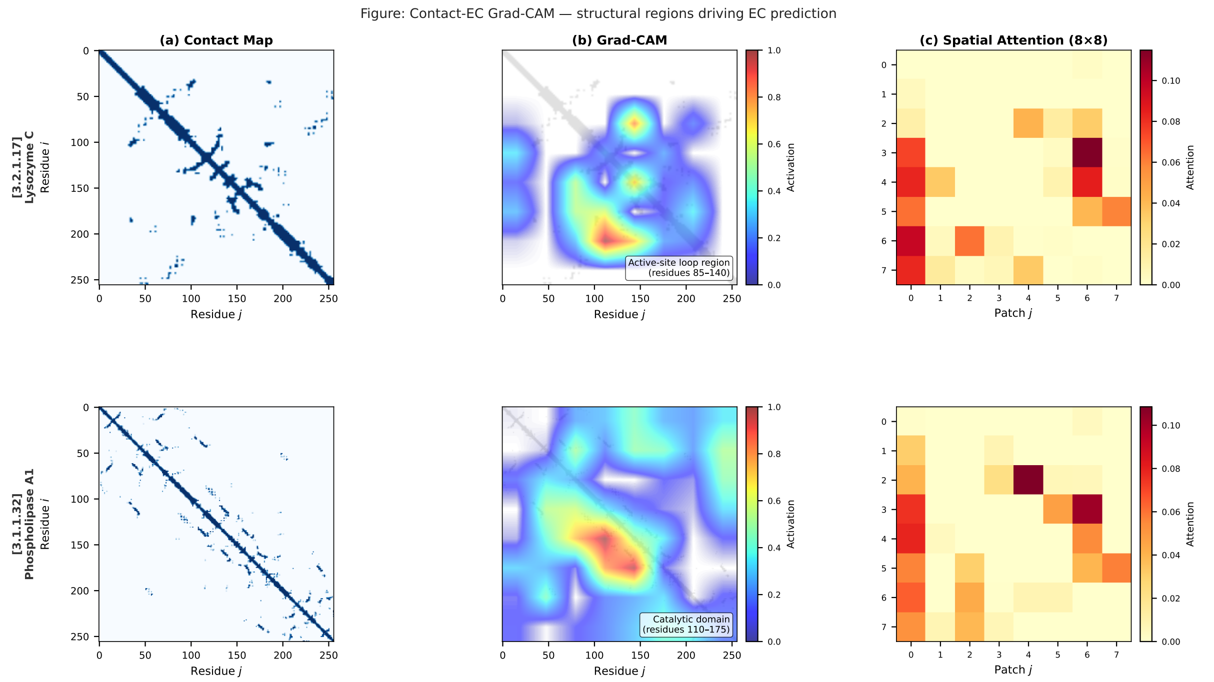

Figure S7: **Grad-CAM structural interpretability for Contact-EC.** Two correctly predicted test-set enzymes. **(a)** AlphaFold-derived contact map. **(b)** Grad-CAM heatmap overlaid on the contact map. **(c)** ResNet-50 spatial attention weights at the 8×8 feature-map level. *Top:* Lysozyme C (EC 3.2.1.17, UniProt C0HLB7). *Bottom:* Phospholipase A1 (EC 3.1.1.32, UniProt P0DPT0).

S8 Extended Discussion

Comparison to CLEAN-Contact. CLEAN-Contact uses ESM-2 3B (2,560-dimensional) with contrastive kNN inference and full Swiss-Prot training data. On New-392, Contact-EC (0.3717, SP-2018) trails CLEAN-Contact (0.566) due to backbone scale (3B vs. 650M) and training recency. Contact-EC-ExpA (SP-2026, 0.7680) surpasses CLEAN-Contact but with potential train/test overlap on New-392.

On the Swiss-Prot 2023 temporal audit, ExpA is instead evaluated on the $N = 99$ encoder-evaluable intersection subset, where it improves over the matched SP-2018 Contact-EC baseline by +9.5 pp micro F1 across three seeds. The gap on Price-149 reflects the OOD nature of bacterial RefSeq sequences and exact Level-4 calibration under structure-source shift rather than a simple missing-input problem.

Paper significance and scope. The current results demonstrate that structural contact maps provide complementary and synergistic information alongside PLM embeddings, empirically supported on rigorous OOD evaluation. The structural parity finding (B3 $\approx$ B1 on hard OOD validation) is, to our knowledge, a novel empirical observation: prior structure-aware EC methods either use contact maps with much larger sequence models or treat structure as a minor addition.

Threshold calibration and temporal generalization. At inference, we apply a fixed sigmoid threshold of 0.5 for Level-4 EC predictions. The sensitivity analysis in Table S7 indicates that this cutoff is reasonably stable for micro F1 but trades precision against recall. The gap in Rare-EC F1 between Contact-EC (0.4219) and Contact-EC-Hier (0.3621) further suggests that the hierarchical head suppresses rare-class outputs through cross-level attention. Future work should explore class-adaptive thresholding and calibrated decision rules to improve rare-class recall without increasing false positive assignments.

3B backbone scaling. Contact-EC-3B (ESM-2 3B backbone, flat FC head, Phase 1 only) achieves micro F1 = 0.6316 on the temporal test, +2.8 pp over the 650M model. The modest gain for a 4.5$\times$ parameter increase suggests the model is data-constrained rather than backbone-constrained at the current training scale.

S9 Reproducibility and Audit Checks

All reported splits were audited for accession overlap, exact sequence overlap, label coverage, multi-label cardinality, and contact-map availability. The audit scripts generate machine-readable CSV summaries used to construct the final result tables. Random-split, cluster-split, EC-Bench-style, and external benchmark results are labeled separately to avoid mixing evaluation protocols.

Sequence-similarity stratification of temporal cases. To test whether temporal performance is concentrated in proteins with close training-set neighbours, we joined the final case-wise temporal comparison ($N = 124$) with the available EC-Bench test-vs-train similarity audit. This similarity file covers 101 of

Table S14: Final temporal leakage and label-encoder audit for the complete-known Swiss-Prot 2023-01 proteins. Accession and exact-sequence overlap are necessary checks but do not establish homolog- or fold-disjointness. ExpA contains many temporal proteins in the full corpus because it is a later Swiss-Prot snapshot, but the training split excludes them; only 99 temporal proteins are Level-4 evaluable under the ExpA encoder.

Training corpus	Train N	Eval. N	Corpus acc. ovl.	Train acc. ovl.	Train seq. ovl.	Encoder L4	Train L4
SP-2018 EC-Bench train	201,865	124	0	0	0	4,647	4,521
SP-2022 train	243,335	124	0	0	0	5,306	5,173
ExpA recent Swiss-Prot train	243,198	99	115	0	0	5,306	5,164

Table S15: Matched temporal recency comparison on the Level-4 encoder-evaluable intersection subset ($N = 99$). This avoids comparing ExpA against the broader SP-2018-only $N = 124$ denominator.

Model	Runs	Micro F1	Weighted F1	Precision	Recall
Contact-EC SP-2018	3 seeds	0.6467 ± 0.0210	0.5692 ± 0.0214	0.7873 ± 0.0175	0.5489 ± 0.0233
Contact-EC-ExpA SP-2026	3 seeds	0.7417 ± 0.0182	0.6768 ± 0.0168	0.8583 ± 0.0096	0.6532 ± 0.0233

Table S16: MMseqs2 nearest-neighbour and label-coverage audit for the temporal recency comparison. Searches use complete Level-4 EC-labelled proteins in each training split as the target database. The newer corpora increase nearest-neighbour identity and top-hit exact EC agreement on the matched $N = 99$ subset, so the ExpA gain should be interpreted as a recent-corpus effect that includes homolog availability and label-frequency changes rather than a pure calendar-date effect.

Subset	Training corpus	N	Median top identity	≥ 0.60 identity	≥ 0.90 identity	All L4 in train	Top-hit same L4
Complete-known	SP-2018	124	0.555	46	9	113	85
Complete-known	SP-2022	124	0.644	63	18	115	95
Complete-known	SP-2026 ExpA	124	0.644	63	18	114	95
Matched intersection	SP-2018	99	0.556	38	8	99	69
Matched intersection	SP-2022	99	0.619	49	14	99	78
Matched intersection	SP-2026 ExpA	99	0.619	49	14	99	78

the 124 fully evaluable temporal proteins; the remaining 23 proteins are therefore excluded from the stratified analysis. Table S17 and Figure S8 show that Contact-EC performance is substantially lower when no recorded sequence similarity to the training set is available, and increases in the 0.60–0.90 identity regime. This supports the manuscript's conservative interpretation: temporal EC performance is sensitive to train-test relatedness, and sequence-disjoint validation does not by itself establish fold-disjoint structural generalization.

Table S17: Temporal performance stratified by maximum recorded train-set sequence identity for the 101 temporal proteins with available EC-Bench similarity metadata. This is a sequence-similarity sensitivity audit, not a structural fold-disjoint evaluation.

Max train identity	N	Contact-EC F1	HIT-EC F1	Contact hit	HIT-EC hit
0.00	32	0.3673	0.8125	0.2812	0.8125
(0.30, 0.60]	36	0.5902	0.8108	0.5000	0.8333
(0.60, 0.90]	27	0.8889	0.9643	0.8889	1.0000
>0.90	6	0.8000	0.8333	0.6667	0.8333

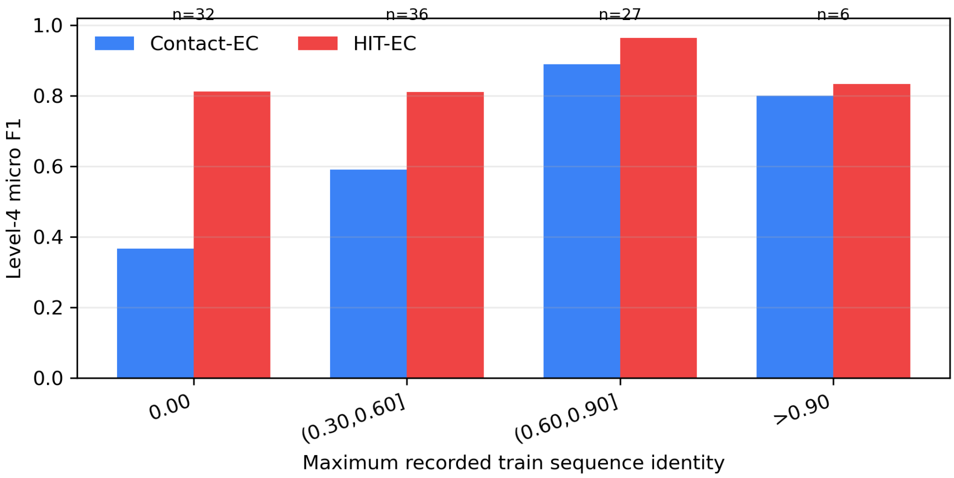

Figure S8: Level-4 micro F1 by maximum recorded train-set sequence identity on the 101 temporal proteins with available EC-Bench similarity metadata. The result motivates the explicit Foldseek-disjoint benchmark reported below.

Contact-pair nearest-neighbor audit. Because sequence-disjoint evaluation does not guarantee structural fold-disjointness, we also performed a model-proximal relatedness audit using the existing contact-pair ESM representation. For each protein, we summarized the contact-pair tensor by concatenating the mean and standard deviation across the sampled residue pairs and computed cosine similarity to all processed SP-2018 training proteins with available contact-pair embeddings. This analysis is not a Foldseek, TM-align, CATH, or SCOP fold-disjoint benchmark. Instead, it asks whether temporal test proteins have close neighbours in the same representation space used by the contact branch. Table S18 and Figure S9 show that all 113 evaluable temporal known-EC proteins have a nearest training neighbour with contact-pair cosine similarity above 0.90. The nearest neighbour shares the same Level-4 EC label in 59.3% of cases. We therefore avoid interpreting the temporal results as evidence of fold-disjoint structural generalization; the Foldseek-disjoint benchmark below addresses this question directly by reclustering the training corpus and retraining the models on the resulting split.

Table S18: Contact-pair nearest-neighbor audit for temporal known-EC proteins. Cosine similarity is computed on mean–standard deviation summaries of existing contact-pair ESM tensors. This is a representation-neighbour audit, not a structural fold-disjoint benchmark.

Quantity	Value
Temporal known-EC proteins with contact-pair embeddings	113/124
SP-2018 training proteins with contact-pair embeddings	212,863/224,294
Median nearest-neighbour cosine	0.9852
Mean nearest-neighbour cosine	0.9845
Temporal proteins with nearest cosine >0.90	113/113
Nearest neighbour same Level-1 EC	87.6%
Nearest neighbour same Level-2 EC	82.3%
Nearest neighbour same Level-3 EC	82.3%
Nearest neighbour same Level-4 EC	59.3%

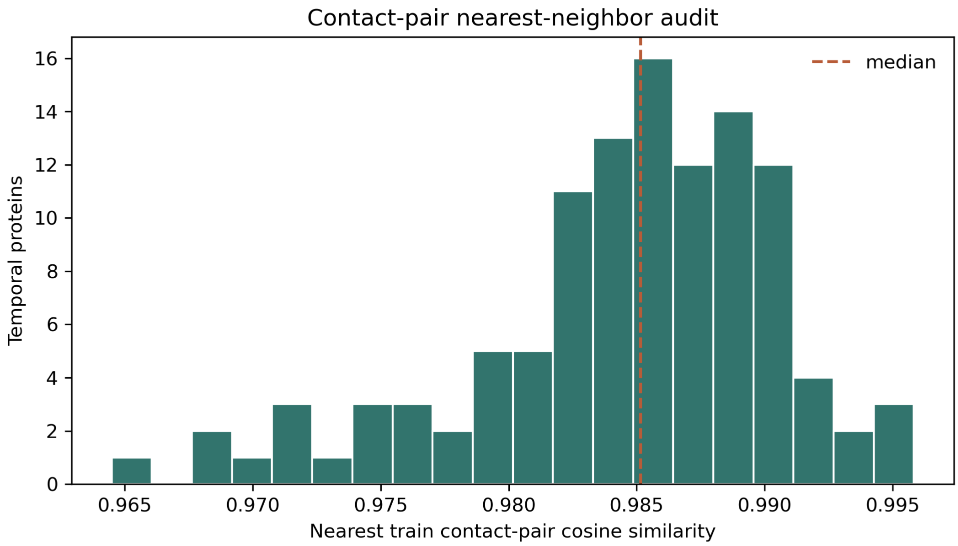

Figure S9: Distribution of nearest SP-2018 training-set cosine similarity for the 113 temporal known-EC proteins with available contact-pair embeddings. The high nearest-neighbour similarities indicate that the current temporal evaluation should not be treated as fold-disjoint structural generalization.

Raw contact-map topology nearest-neighbor audit. To make the structural-relatedness audit less dependent on the learned contact-pair ESM representation, we repeated the nearest-neighbor analysis using hand-crafted fingerprints derived directly from the 256×256 contact maps. Each fingerprint concatenates a 32×32 coarse contact grid, sequence-separation contact densities, and node-degree distribution statistics ($d = 1041$). This is still not a Foldseek/TM-align or CATH/SCOP fold-disjoint benchmark, but it uses the raw structural input rather than a model-internal representation. Table S19 and Figure S10 show that 114/115 temporal proteins with available contact maps have a nearest SP-2018 training protein with topology cosine similarity >0.90, and 63/115 exceed 0.98. The nearest topology neighbour shares the same Level-4 EC label in 40.9% of cases, rising to 57.1% among proteins with cosine similarity >0.98. This independent audit reinforces the conservative interpretation: the temporal split measures annotation-time shift and label-vocabulary shift, but it should not be presented as fold-disjoint structural generalization.

Table S19: Raw contact-map topology nearest-neighbor audit for temporal known-EC proteins. Similarity is computed on hand-crafted fingerprints from the 256×256 contact maps, independent of the learned contact-pair ESM representation.

Quantity	Value
Temporal known-EC proteins with contact maps	115/124
SP-2018 training proteins with contact maps	212,863/224,294
Fingerprint dimension	1,041
Median nearest-neighbour cosine	0.9822
Mean nearest-neighbour cosine	0.9733
Temporal proteins with nearest cosine >0.90	114/115
Temporal proteins with nearest cosine >0.95	93/115
Temporal proteins with nearest cosine >0.98	63/115
Nearest neighbour same Level-1 EC	75.7%
Nearest neighbour same Level-2 EC	69.0%
Nearest neighbour same Level-3 EC	69.0%
Nearest neighbour same Level-4 EC	40.9%

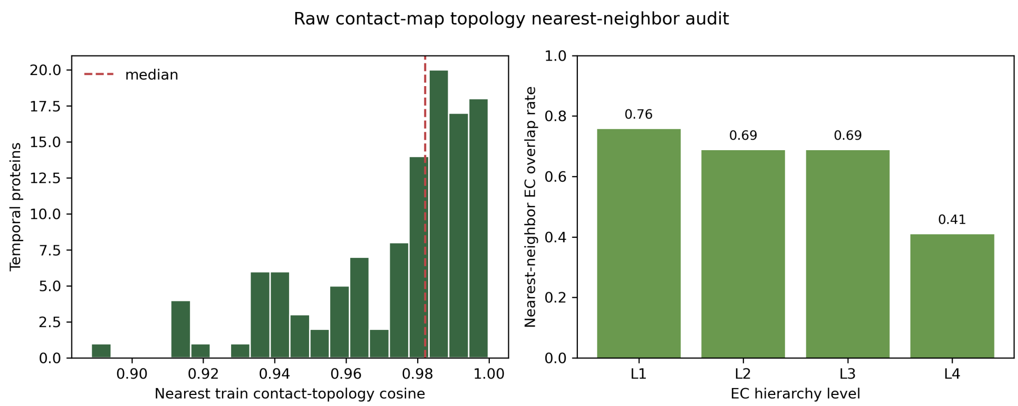

Figure S10: Raw contact-map topology nearest-neighbor audit. Left: nearest SP-2018 training-set cosine similarity for temporal known-EC proteins with available contact maps. Right: EC hierarchy overlap with the nearest topology neighbour.

Foldseek/TM-score-disjoint benchmark. We constructed an explicit fold-level split by clustering all SP-2018 proteins with available AlphaFold structures using Foldseek [15]. The clustering used TM-score threshold 0.50, coverage 0.80, minimum sequence identity 0.0, alignment type 1, cluster mode 1, and sensitivity 7.5. Of 224,294 metadata proteins, 212,863 had available structures and were assigned to 3,813 clusters. The cluster-count-balanced split contains 170,573 train proteins, 25,565 validation proteins, and 16,725 test proteins, with 3,051/381/381 structure clusters and zero protein-ID overlap across partitions.

Table S20: Foldseek/TM-score-disjoint split audit. Missing-structure proteins are listed in the released audit files and excluded from this structure-disjoint training/evaluation split.

Quantity	Value
Metadata proteins	224,294
Proteins with available structures	212,863
Missing structures	11,431
Foldseek clusters	3,813
Cluster size median / mean / max	2.0 / 55.83 / 18,692
Train proteins / clusters	170,573 / 3,051
Validation proteins / clusters	25,565 / 381
Test proteins / clusters	16,725 / 381
Train–validation ID overlap	0
Train–test ID overlap	0
Validation–test ID overlap	0

Table S21: Foldseek/TM-score-disjoint Level-4 performance for the representative base checkpoints. The fixed threshold 0.5 is the default sigmoid cutoff. Validation-selected thresholds are selected on the Foldseek validation split and applied once to the held-out Foldseek test split.

Model	F1@0.5	Val thr.	Test F1@val thr.	Precision	Recall
B1 (ESM-2, flat FC)	0.0452	0.130	0.0613	0.1407	0.0392
B3 (contact only)	0.0436	0.070	0.0652	0.0619	0.0690
Contact-EC (fusion)	**0.0864**	0.080	**0.1747**	**0.2406**	**0.1371**

Table S22: Foldseek/TM-score sweep split composition. The 0.50 split was generated with cluster-count assignment, whereas the 0.40 and 0.60 splits were generated with protein-balanced assignment. Thus the sweep is a robustness/composition audit rather than a strictly monotonic threshold-only curve.

TM-score	Assignment	Clusters	Max cluster	Test proteins	Unseen pos.	Any-unseen proteins
0.40	protein-balanced	10,893	1,026	21,617	0.159	0.170
0.50	cluster-count	3,813	18,692	16,725	0.315	0.316
0.60	protein-balanced	11,869	1,026	21,555	0.103	0.110

The Foldseek split also changes label coverage. Among Foldseek-test positive Level-4 labels, 31.5% are absent from the corresponding Foldseek training split, and 31.6% of test proteins contain at least one unseen Level-4 label. Therefore, fold-disjoint evaluation mixes structural shift with closed-set vocabulary and frequency shift; the latter must be reported rather than attributed solely to fold-level generalization.

These protocol distinctions are used throughout the manuscript to avoid treating all F1 values as interchangeable. In particular, the Bioinformatics-relevant claim is not that Contact-EC is the strongest

Table S23: Single-seed neural model performance across Foldseek/TM-score sweep splits at the fixed Level-4 threshold of 0.5. Fusion is consistently strongest, but absolute scores must be interpreted together with the split-composition audit in Table S22.

TM-score split	B1 ESM-2	B3 contact	Contact-EC fusion
0.40	0.0683	0.2622	**0.2981**
0.50	0.0462	0.0337	**0.0998**
0.60	0.0538	0.3985	**0.4403**

Table S24: MMseqs2 top-hit homology-transfer baseline on Foldseek/TM-score split variants. The baseline transfers all Level-4 EC labels from the single best MMseqs2 hit in the corresponding training partition. Higher TM-score thresholds produce stricter clustering in the newly generated threshold sweep.

Split	Coverage	L1 hit	L2 hit	L3 hit	L4 hit	L4 micro F1
TM-score 0.40	0.902	0.865	0.798	0.766	0.660	0.694
TM-score 0.50	0.753	0.706	0.642	0.614	0.473	0.539
TM-score 0.60	0.962	0.938	0.933	0.933	0.802	0.818

Table S25: MMseqs2 top-hit homology-transfer baseline on temporal and external test sets using the SP-2018 training corpus as the search database. Price-149 is reported both on the encoded subset used by the neural models and on the raw 149-protein benchmark.

Test set	Coverage	L1 hit	L2 hit	L3 hit	L4 hit	L4 micro F1
Swiss-Prot 2023 known EC ($N = 124$)	0.871	0.831	0.831	0.831	0.637	0.585
Price-149 encoded ($N = 136$)	0.963	0.941	0.875	0.809	0.338	0.345
Price-149 raw ($N = 149$)	0.960	0.940	0.872	0.805	0.309	0.312

Table S26: Three-seed temporal known-EC Level-4 performance on the complete Swiss-Prot 2023-01 subset ($N = 124$). Values are mean $\pm$ sample standard deviation across seeds 42/43/44.

Model	Micro F1	Weighted F1	Precision	Recall	Rare-EC F1
B1 (ESM-2, flat FC)	0.4508 ± 0.0203	0.3387 ± 0.0194	$\mathbf{0.9035 \pm 0.0445}$	0.3006 ± 0.0173	0.3292 ± 0.0272
B3 (contact only)	0.4244 ± 0.0207	0.3325 ± 0.0237	0.8300 ± 0.0220	0.2854 ± 0.0200	0.3354 ± 0.0254
Contact-EC flat FC	$\mathbf{0.6241 \pm 0.0170}$	$\mathbf{0.5278 \pm 0.0222}$	0.7997 ± 0.0226	$\mathbf{0.5120 \pm 0.0200}$	$\mathbf{0.5257 \pm 0.0313}$

Table S27: MMseqs2 top-hit performance by percent-identity bin on the primary TM-score 0.50 Foldseek split. This analysis shows that Foldseek-disjoint evaluation still contains substantial sequence-homology signal for many test proteins.

Top-hit identity bin	Proteins	Fraction	L4 hit	L4 micro F1
No hit	4,138	0.247	0.000	0.000
0–20%	480	0.029	0.250	0.250
20–30%	3,748	0.224	0.466	0.466
30–50%	6,849	0.410	0.677	0.677
50–70%	857	0.051	0.903	0.903
$\geq$70%	653	0.039	0.960	0.960

Table S28: Three-seed Foldseek/TM-score-disjoint Level-4 performance. Values are mean $\pm$ sample standard deviation over the base checkpoint and two additional training seeds. Validation thresholds are selected on the Foldseek validation split and applied once to the held-out Foldseek test split.

Model	F1@0.5	Test F1@val thr.	Post-hoc best F1
B1 (ESM-2, flat FC)	0.0433 ± 0.0042	0.0607 ± 0.0009	0.0666 ± 0.0042
B3 (contact only)	0.0400 ± 0.0054	0.0654 ± 0.0016	0.0700 ± 0.0036
Contact-EC (fusion)	$\mathbf{0.0872 \pm 0.0122}$	$\mathbf{0.1685 \pm 0.0334}$	$\mathbf{0.1769 \pm 0.0378}$

Table S29: Three-seed Foldseek-disjoint top-k diagnostics. Hit rate counts whether any true Level-4 EC label appears among the top-k predicted labels; values are mean $\pm$ sample standard deviation.

Model	Top-1 hit	Top-3 hit	Top-5 hit	Top-10 hit
B1 (ESM-2, flat FC)	0.0503 ± 0.0013	0.0977 ± 0.0029	0.1203 ± 0.0015	0.1455 ± 0.0043
B3 (contact only)	0.0583 ± 0.0089	0.1105 ± 0.0051	0.1317 ± 0.0102	0.1566 ± 0.0093
Contact-EC (fusion)	$\mathbf{0.1383 \pm 0.0381}$	$\mathbf{0.2445 \pm 0.0489}$	$\mathbf{0.2778 \pm 0.0500}$	$\mathbf{0.3257 \pm 0.0434}$

Table S30: Foldseek-disjoint failure-mode audit across three seeds. A rescue means that fusion predicts at least one true Level-4 EC label for a protein for which the comparison baseline has zero sample-level hit rate across seeds.

Quantity	Fraction of Foldseek-test proteins
Fusion rescue vs. B1	0.173
Fusion rescue vs. B3	0.125
Fusion-only rescue vs. both B1 and B3	0.110
Missed by all three models	0.753
All-model failures with any unseen true label	0.419
Fusion-only rescues with any unseen true label	0.005

Table S31: Representative Foldseek-disjoint fusion-only rescue cases. These examples have seen true Level-4 labels in the Foldseek training partition, zero hit rate for both single-modality baselines across seeds, and nonzero fusion hit rate.

Accession	EC family	True EC	Frequency bin	Train count	Fusion hit	Fusion top-1
P43418	Hydrolases	3.1.21.4	26–100	95	1.000	1.000
A5F3I6	Oxidoreductases	1.8.4.8	6–25	8	1.000	1.000
P51040	Transferases	2.3.3.16	6–25	9	1.000	1.000
Q9CB92	Lyases	4.2.99.18	26–100	54	0.667	0.667
Q9ZFH0	Isomerases	5.1.3.37	6–25	7	0.667	1.000

Table S32: Foldseek-disjoint open-vocabulary confidence audit. The target is whether a test protein contains at least one true Level-4 label absent from the Foldseek training split. Scores are unsupervised confidence diagnostics averaged over three seeds; higher AUROC/AP indicates better detection of unseen-label cases.

Model	Best confidence score	AUROC	AP
B1 (ESM-2, flat FC)	Low top-5 probability sum	0.676	0.436
B3 (contact only)	Low top-1 probability	0.574	0.365
Contact-EC (fusion)	Low top-5 probability sum	**0.738**	**0.590**

Table S33: Foldseek-disjoint abstention and parent-level fallback audit for three-seed averaged Contact-EC fusion probabilities. Proteins are abstained in ascending order of top-5 Level-4 probability sum. Retained Level-4 micro F1 is computed only on accepted predictions; abstained Level-3 top-1 hit evaluates a conservative parent-level fallback for deferred proteins.

Abstained	Coverage	Retained L4 micro F1	Abstained unseen prevalence	Abstained L3 top-1 hit
0%	1.00	0.208	–	–
10%	0.90	0.222	0.712	0.059
20%	0.80	0.239	0.592	0.112
30%	0.70	0.257	0.533	0.157
40%	0.60	0.279	0.496	0.185
50%	0.50	0.305	0.465	0.204

Table S34: Foldseek-disjoint label-frequency decomposition. Values are Level-4 micro F1 at model-specific validation-selected thresholds, stratified by the number of Foldseek-train proteins carrying each true Level-4 label. These values come from the canonical checkpoint outputs used for the label-frequency audit, whereas Table S28 reports three-seed means for the primary all-label metric.

Model	unseen	1–5	6–25	26–100	>100
B1 (ESM-2, flat FC)	0.0000	0.0000	0.0000	0.0847	0.1242
B3 (contact only)	0.0000	0.0055	0.0448	0.1081	0.1825
Contact-EC (fusion)	0.0000	0.0038	**0.2484**	**0.3278**	**0.2167**

Table S35: Foldseek-disjoint aggregate performance after excluding unseen positive Level-4 labels from the same canonical prediction outputs used in Table S34. Seen-label-only aggregation removes the impossible closed-set positives from the denominator and shows that fusion retains a larger gain over both single-modality baselines.

Model	Positive labels	Predictions	Precision	Recall	Micro F1
B1 (ESM-2, flat FC)	11,622	13,866	0.0793	0.0946	0.0863
B3 (contact only)	11,622	10,511	0.0905	0.0818	0.0859
Contact-EC (fusion)	11,622	13,370	0.2018	0.2321	**0.2159**

Table S36: Foldseek-disjoint sample-level hit rate by true-label training frequency. Values are averaged over three seeds using validation-selected thresholds. The fusion gain is concentrated in seen labels, especially 6–100 training examples, whereas unseen labels remain essentially unsolved by closed-set models.

True-label bin	N	Any unseen	B1 hit	B3 hit	Fusion hit
Unseen	5,287	1.000	0.000	0.000	0.002
1–5	4,025	0.000	0.000	0.020	0.031
6–25	2,415	0.000	0.000	0.047	0.187
26–100	3,243	0.000	0.001	0.157	0.193
>100	1,755	0.000	0.376	0.275	0.537

Table S37: Foldseek-disjoint EC Level-1 family performance at model-specific validation-selected thresholds.

Family	B1 F1	B3 F1	Fusion F1
Oxidoreductases	0.1157	0.0507	**0.2612**
Transferases	0.0635	0.0606	**0.1497**
Hydrolases	0.0777	0.0967	**0.2576**
Lyases	**0.1288**	0.1135	0.0468
Isomerases	**0.2468**	0.2440	0.2318
Ligases	0.0011	0.0005	**0.0036**

Table S38: Interpretation of each evaluation protocol. The same numerical metric can answer different scientific questions depending on split construction and label coverage.

Protocol	**Primary question**	**Main caveat**
Random Swiss-Prot split	In-distribution fitting and calibration	Contains exact sequence duplicates across partitions; not evidence of temporal or sequence-disjoint generalization.
MMSeqs2 cluster split	Generalization to sequence-disjoint proteins	Valid only when the evaluated checkpoint is trained on the corresponding cluster-train split.
EC-Bench val_hard	Controlled hard validation at $\leq$30% sequence identity	Still a validation protocol; temporal deployment claims require held-out future annotations.
Swiss-Prot 2023-01 known-EC subset	Closed-set recognition of temporally new proteins whose labels are in vocabulary	Only 124/468 proteins are fully evaluable at Level 4; contact-pair and raw contact-map audits show high nearest-neighbour structural relatedness.
Swiss-Prot 2023-01 full set	Sensitivity to temporal label shift and incomplete annotations	Operational coverage audit only; partial labels are not fully Level-4 scorable, and novel labels are outside the closed-set vocabulary.
New-392 / Price-149	External stress tests from CLEAN-Contact	New-392 overlaps with current Swiss-Prot; Price-149 is an external bacterial RefSeq stress test with strong Level-4 specificity loss despite available inputs.
Foldseek/TM-score-disjoint split	Closed-set EC prediction under explicit structure-cluster shift	Absolute Level-4 performance is low and label coverage varies across generated TM-score sweep splits; the primary 0.50 split has 31.5% unseen positive labels, so results must be interpreted jointly with label coverage and split composition.

available EC predictor in absolute terms. Rather, Contact-EC provides a controlled sequence–structure framework for separating representation effects from temporal cutoff, label-vocabulary shift, sequence overlap, and structure availability.

Canonical result selection. When multiple evaluation logs exist for the same checkpoint, we use the full multi-label evaluation on the full evaluable temporal set ($N = 124$) as canonical. Earlier $N = 101$ evaluations are retained only as historical partial runs and are not used in the main conclusions. For Contact-EC-3B, the canonical temporal result is micro F1 = 0.6316 from the full $N = 124$ multi-label evaluation; auxiliary argmax-only and partial-set evaluations are excluded from the main result table. For B3 contact-only hard validation, we use the completed resumed checkpoint evaluation (micro F1 = 0.7650, weighted F1 = 0.6788, macro F1 = 0.1335); an earlier interrupted training log reached a higher validation micro F1 before termination but is not used as the paper source because the completed run provides the auditable checkpoint/result pair.

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